

NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction

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Abstract—Selection-based coverless communication requires verifying that a finite image index can satisfy every message symbol *before* transmission, not merely that a descriptor achieves high average robustness. This paper presents NARCIS (Neural Adaptive Robust Coverless Image Signaling), an end-to-end protocol that transmits unchanged natural images while providing authenticated payload protection, replay rejection, and forward error correction. A compact self-supervised encoder trained with a variance-invariance-covariance objective maps images to normalised 64-dimensional descriptors. A quantile codebook partitions the principal descriptor direction into balanced symbol classes. A multi-attack qualification stage retains only covers whose labels remain stable under all calibration conditions. Payloads are protected with AES-GCM; symbols are assigned through a keyed Gray permutation; Reed-Solomon coding corrects residual label errors.

Evaluation uses five deterministic partitions each of BOSSBase-1.01 (grayscale, 512×512) and Caltech-101 (variable-resolution RGB). Across 1,575 complete-message trials under clean, calibration, and holdout conditions, NARCIS recovered every authenticated plaintext. Mean symbol accuracies were 98.74% and 98.59%. Net throughput at $K = 16$ scales from 0.184 bits per cover for 8-byte payloads to 1.213 bits per cover for 128-byte payloads. Removing attack qualification on BOSSBase seed 11 reduced complete-message success from 210/210 to 142/210 (67.6%), while a calibration-locked projection recovered 210/210. Selection bias was measured with three detector classes. On BOSSBase, all confidence intervals include chance performance. On Caltech-101, classical detectors identified a weak selection shift (global logistic AUC 0.531, 95% CI [0.504, 0.557]; residual ExtraTrees 0.533, [0.504, 0.562]); 20 repeated CNN fits on the most sensitive partition confirmed a partition-specific signal (mean AUC 0.533, 95% CI [0.517, 0.548]).

Index Terms—Coverless image steganography, neural image descriptor, robust codebook, authenticated encryption, error

correction, cover selection, Reed-Solomon, AES-GCM.

I. INTRODUCTION

SLECTION-BASED coverless image steganography communicates information by choosing natural images from a shared index rather than by modifying a carrier. The transmitted objects are unaltered photographs. Three interconnected gaps limit this approach's operational viability.

Gap 1: Finite-index message feasibility is not verified before transmission. Existing methods report nominal hash length or average symbol accuracy. Neither metric guarantees that the required sequence of symbols can be satisfied by the available, qualified index for a given message. A transmission that cannot be realised fails silently rather than signalling infeasibility, producing message-dependent coverage gaps that are invisible in per-symbol statistics [1], [2].

Gap 2: No end-to-end coverless protocol integrates authentication, replay control, and error correction. Selection-based schemes propose that cover integrity replaces steganographic hiding, but they leave payload confidentiality, metadata authenticity, replay rejection, and channel error recovery as open problems. The published literature treats these components individually or omits them entirely within the coverless framework [3]–[6].

Gap 3: Cover stability filtering creates unquantified selection bias. Retaining only stable covers changes the distribution of transmitted images relative to the full index. Whether this selection leaks detectable information depends on the dataset, the filtering policy, and the detector, but this three-way dependence is rarely measured [7], [8].

NARCIS (Neural Adaptive Robust Coverless Image Signaling) closes all three gaps simultaneously.

Contribution 1: Feasible operating point. A codebook is accepted only when every attack-qualified bucket satisfies the multiplicity required by each protected coded message. The selected operating point is the largest alphabet consistent with this criterion. Across 1,575 end-to-end trials on two datasets and 21 channel conditions, NARCIS recovered 100% of authenticated plaintexts.

Contribution 2: End-to-end cryptographic and coding integration. AES-GCM [9] authenticates payload and session metadata with associated sender-receiver identities; a keyed Gray permutation limits public symbol-to-label correspondence; monotonic sequence checking rejects replay;

This research received no external funding. The authors declare no competing interests. WYSAWIS, cited in the related work, shares authors with this study and is explicitly identified as such; it is used as published qualitative context, not as an experimental baseline. (Corresponding author: Stéphane Gaël R. Ekodeck, stephane-gael.ekodeck@facsclences-uy1.cm.)

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Reed-Solomon coding [10] corrects residual label errors. Net throughput at $K=16$ scales from 0.184 to 1.213 bits per cover for 8 to 128-byte payloads (Section VI-C).

Contribution 3: Systematic detectability measurement.

Three detector classes (global logistic regression, residual feature ensemble, and a selection CNN) are applied to five partitions of each dataset. On BOSSBase all confidence intervals include chance performance. Caltech-101 shows a partition-specific CNN signal (mean AUC 0.533, 95% CI [0.517, 0.548] over 20 repeated fits on the most sensitive partition), reported transparently rather than suppressed (Section VI-E).

The remainder of the paper is organised as follows. Section II reviews related work. Section III states the system and threat model with formal security propositions. Section IV presents the NARCIS components. Section V describes the experimental protocol. Section VI reports all results. Section VII interprets them. Section VIII concludes.

II. RELATED WORK

A. Hash- and retrieval-based coverless communication

Early coverless systems mapped robust image hashes to message fragments [1], [3]. Later retrieval-based schemes used local descriptors to improve resistance to common processing [4]. These methods preserve the transmitted image, but the database must contain sufficiently many stable representatives for every message symbol.

Abdulsattar [11] increased single-image nominal capacity by dividing a grayscale cover into overlapping blocks and mapping eigenvalue comparisons to ASCII values. WYSAWIS [12] groups 15-bit block hashes into 16 sequence categories and uses cloud-file rankings to represent segments. Both works illustrate that a large nominal representation space does not by itself guarantee per-message feasibility under finite-index constraints.

B. Learned, joint, and stability-aware methods

Neural feature mapping and generation provide semantic representations or larger message spaces [2], [13]. Selection and generation have been combined with StarGAN [14], while information-driven generation has been applied to control message-to-content mappings [15]. Recent semantic approaches derive symbols from neglected objects, human poses, or detections [16]–[18]. Other systems use semantic text-to-image generation or latent diffusion to enlarge the generative cover space [19], [20].

Ren and Wu [21] combined selection and generation modules to address mapping completeness and reconstruction. Guo et al. [5] studied robustness and diversity against passive and active analysis. Guo and Ping [6] used pseudo-Zernike moments, stability-aware quantisation, and stability-regularised clustering, reporting average robustness of 99.54%, 98.64%, and 97.19% at 10, 14, and 15 bits on Holidays, VOC, and ImageNet, respectively. These results use different datasets, attack sets, and protocol overheads; direct numerical comparison with NARCIS is not attempted, and the cited figures serve as qualitative context.

NARCIS shares the stability-aware objective but focuses on protocol completeness rather than descriptor-level capacity: its primary evaluation unit is complete authenticated message recovery under a finite qualified index.

C. Security and evaluation

No pixel modification removes embedding artefacts, but it does not establish communication security. Selection may alter the distribution of transmitted content, and metadata can reveal protocol state. Steganalysis research has shown that weak statistical differences can be exploitable [7], [22]–[26]. Formal steganographic security distinguishes distributional secrecy from cryptographic payload protection [8]. NARCIS therefore separates cryptographic mechanisms (Propositions 1 and 2) from empirical selection detectability (Section VI-E).

Table I separates existing families by cover formation, mapping mechanism, and protocol-level protection.

III. SYSTEM AND THREAT MODEL

A. Shared index and protocol parties

The sender and receiver share an indexed corpus $\mathcal{I} = \{(x_i, y_i)\}_{i=1}^N$, a symbol-assignment key k_s , a payload encryption key k_p , and a metadata seal key k_m . Each x_i is a natural image and $y_i \in \{0, \dots, K-1\}$ is its codebook label. The sender transmits a sequence of unchanged images; the receiver applies the same descriptor and codebook to recover labels and decode the protected payload.

B. Channel model

The channel may apply JPEG compression, additive Gaussian noise, Gaussian blur, resizing, central cropping, or small rotations. The declared channel model determines which calibration transformations are used to qualify covers and train the descriptor.

C. Protocol targets

The protocol enforces five properties: (i) *cover integrity*: no pixel modification by the sender; (ii) *message correctness*: exact plaintext recovery or explicit failure; (iii) *confidentiality and authenticity*: authenticated encryption of payload and metadata; (iv) *freshness*: rejection of replayed or reordered sequences; (v) *channel robustness*: correction of residual label errors within the Reed-Solomon budget.

D. Formal security properties

The following propositions state the cryptographic guarantees provided by NARCIS under standard assumptions. They follow directly from the authenticated encryption security of AES-GCM [9] and require no new cryptographic proof.

Proposition 1 (Payload Confidentiality): For plaintext m , nonce derived from sequence number n , and key k_p unknown to the adversary, the ciphertext $c = \text{AEAD.Enc}_{k_p}(m; \text{nonce}, \text{AD})$ is computationally indistinguishable from a uniformly random string of the same length,

TABLE I

STRUCTURAL COMPARISON OF COVERLESS IMAGE-STEGANOGRAPHY FAMILIES. “PROTOCOL PROTECTION” DENOTES INTEGRATED AUTHENTICATED PAYLOAD HANDLING AND ERROR CORRECTION RATHER THAN IMAGE-LEVEL ROBUSTNESS ALONE.

Family	Representative works	Cover formation	Mapping / descriptor	Protocol protection
Block/hash mapping	Abdulsattar [11]; WYSAWIS [12]	Existing natural images	Block comparisons or category hashes	Not central to the cited mapping mechanism
Retrieval and local features	Li et al. [4]	Selected natural images	SURF-based retrieval	Not central to the cited retrieval mechanism
Adversarial generation	Qin et al. [13]; Chen et al. [14]; Zhang et al. [15]	Generated or selected/generated images	Latent or information-driven neural mapping	Evaluated mainly at image/message mapping level
Semantic mapping	Zou et al. [16]; Tan et al. [17]; Meng et al. [18]	Existing natural images	Objects, poses, or detector outputs	Not the primary contribution
Text-to-image and diffusion	Li et al. [19]; Guo et al. [20]	Generated images	Semantic prompts or latent transformations	Not directly comparable to finite-index retrieval
Stability-aware selection	Guo et al. [5]; Guo-Ping [6]	Selected natural images	Stable handcrafted features and clustering	Image-level robustness and diversity
NARCIS	This work	Selected unchanged natural images	Invariant neural descriptor and qualified quantile codebook	AES-GCM, replay control, Reed-Solomon, exact recovery

with advantage bounded by the IND-CCA2 security of AES-GCM [9]. An adversary observing transmitted images and sealed metadata gains no information about m .

Proposition 2 (Authenticity and Replay Rejection): A message is accepted by the receiver only when (i) the AES-GCM tag on the sealed metadata verifies under k_m with sender-receiver identities as associated data, and (ii) the sequence number n embedded in the metadata satisfies $n > n_{\max}$, where $n_{\max}$ is the highest previously accepted value for that sender. Under AES-GCM’s authentication guarantee [9], any altered or replayed message fails condition (i) or condition (ii).

Traffic analysis and semantic inference from image content remain outside the cryptographic claim. Selection detectability is treated as a measured empirical property (Section VI-E).

IV. NARCIS METHOD

A. Channel-invariant neural descriptor

Channel transformations are applied at the declared channel resolution, and the resulting images are resized to 128×128 for descriptor inference. The encoder has four convolutional blocks with 16, 32, 64, and 96 channels; each block uses two 3×3 convolutions, batch normalisation, and SiLU activations; the last three blocks downsample by a factor of two. Global average pooling and a two-layer $96 \rightarrow 128 \rightarrow 64$ projection head produce a 64-dimensional unit-normalised descriptor $z = f_{\theta}(x)$. The grayscale and RGB instances have 231,312 and 231,600 trainable parameters, respectively; only the first convolution differs. Fig. 1 shows the implemented dimensions.

Training uses two independently distorted views $x^{(1)}$ and $x^{(2)}$ of the same image. Each view is generated by randomly selecting JPEG compression, Gaussian noise, blur, resize, crop, rotation, or the identity transform. The loss is

$$\mathcal{L} = \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cov}}, \quad (1)$$

where $\mathcal{L}_{\text{inv}}$ is the mean squared distance between paired descriptors, $\mathcal{L}_{\text{var}}$ prevents dimensional collapse, and $\mathcal{L}_{\text{cov}}$

penalises off-diagonal covariance. This variance-invariance-covariance structure follows the non-contrastive principle of VICReg [27]. The weights are $(\lambda_{\text{inv}}, \lambda_{\text{var}}, \lambda_{\text{cov}}) = (25, 25, 1)$; the target standard deviation is $64^{-1/2}$, consistent with unit-normalised descriptors.

For a batch of B paired d -dimensional descriptors Z and Z' :

$$\mathcal{L}_{\text{inv}} = \frac{1}{Bd} \|Z - Z'\|_F^2, \quad (2)$$

$$\mathcal{L}_{\text{var}} = \frac{1}{d} \sum_{j=1}^d [\max(0, \gamma - \sigma_j(Z)) + \max(0, \gamma - \sigma_j(Z'))], \quad (3)$$

$$\mathcal{L}_{\text{cov}} = \frac{1}{d(d-1)} \sum_{i \neq j} (C_{ij}(Z)^2 + C_{ij}(Z')^2), \quad (4)$$

where $\gamma = d^{-1/2}$, σ_j is the batch standard deviation with 10^{-4} variance regularisation, and $C(Z) = (Z - \bar{Z})^\top (Z - \bar{Z}) / (B - 1)$.

B. Balanced quantile codebook

Let $Z \in \mathbb{R}^{N \times 64}$ contain clean descriptors and let v_1 be the first right singular vector of the centred matrix. Each descriptor is projected as

$$s_i = (z_i - \bar{z})^\top v_1. \quad (5)$$

For a candidate alphabet of size K , boundaries are placed at the empirical quantiles j/K , $j = 1, \dots, K-1$. The resulting label is

$$q_K(z_i) = \sum_{j=1}^{K-1} \mathbf{1}[s_i > b_j]. \quad (6)$$

Quantile partitioning balances clean occupancy before channel qualification and avoids empty clusters from unconstrained fitting. In the reduced sensitivity experiment, PC1 explained 72.90% of descriptor variance; Section VI-F analyses alternative projections.

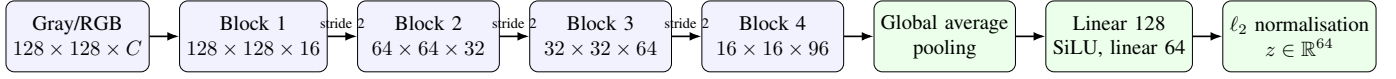


Fig. 1. Descriptor encoder used in both principal evaluations, with $C=1$ for BOSSBase and $C=3$ for Caltech-101. Each convolutional block contains two 3×3 convolution-batch-normalisation-SiLU stages; channels transformations precede model resizing to 128×128 .

C. Attack-qualified cover index

A cover is retained only when its label is unchanged under every calibration condition:

$$\mathcal{S}_K = \{i : q_K(f_\theta(T(x_i))) = q_K(f_\theta(x_i)), \forall T \in \mathcal{T}_{\text{cal}}\}. \quad (7)$$

The calibration set $\mathcal{T}_{\text{cal}}$ contains 12 transformations: JPEG qualities 80 and 50; Gaussian noise standard deviations 5 and 12; blur radii 0.8 and 1.5; resize factors 0.75 and 0.50; crop fractions 0.05 and 0.10; rotations of 3° and 7° .

Seven image statistics are used to estimate selection bias: mean, standard deviation, horizontal and vertical gradient magnitude, entropy, and the 0.1 and 0.9 intensity quantiles. Within each symbol bucket, stable covers are ordered by clipped inverse-propensity weighted random priorities. This does not change labels or pixels; it reduces deterministic preference for visually distinctive stable covers.

D. Protected message representation

For plaintext m and sequence number n , AES-GCM produces

$$c = \text{AEAD.Enc}_{k_p}(m; \text{nonce}, \text{NARCIS-PAYLOAD-1} \| n). \quad (8)$$

The ciphertext envelope is framed with a protocol marker, length field, and CRC-32, then encoded with Reed-Solomon parity [10]. Session metadata contains the sequence number, padding, codebook size, and cover count; it is independently sealed with AES-GCM using sender and receiver identities as associated data. The receiver accepts only a sequence number greater than the highest previously accepted value for that sender (Proposition 2).

For $K = 2^b$, the coded bit stream is divided into b -bit symbols. A keyed permutation combines a Gray ordering, a key-derived cyclic shift, and a key-derived orientation. Let $h = \text{HMAC-SHA256}(k_s, \text{cluster-order})$ [28], $a = \text{int}(h_{0:8}) \bmod K$, and $r = -1$ if the least significant bit of h_8 is one, otherwise $r = 1$. For position $p \in \{0, \dots, K-1\}$, define the binary-reflected Gray symbol $g(p) = p \oplus (p \gg 1)$ and

$$\pi_k(g(p)) = (a + rp) \bmod K. \quad (9)$$

Adjacent symbol values remain adjacent in the unshifted codebook order, while the externally observed symbol-to-label assignment depends on the key.

E. Feasible operating point

Let $d_k(c)$ denote the number of occurrences of label k required to transmit envelope c , and let a_k be the number of unused qualified covers in bucket k . A candidate codebook is feasible when

$$d_k(c) \leq a_k, \quad \forall k, \forall c \in \mathcal{C}_{\text{eval}}. \quad (10)$$

Algorithm 1: NARCIS protected coverless transmission

Input: Plaintext m , sequence n , qualified index $\mathcal{I}$, keys

Output: Recovered plaintext or failure

- 1 Encrypt m with sequence-bound AES-GCM;
 - 2 Frame the ciphertext and apply Reed-Solomon coding;
 - 3 Select the largest feasible codebook K ;
 - 4 Map coded symbols to labels with the keyed permutation π_k ;
 - 5 Select one unused qualified cover for each required label;
 - 6 Transmit the unchanged cover sequence and sealed metadata;
 - 7 Verify metadata and reject replayed sequence numbers;
 - 8 Classify received covers and invert the keyed mapping;
 - 9 Apply Reed-Solomon decoding and verify CRC;
 - 10 Authenticate and decrypt the payload;
 - 11 **return** plaintext if every verification succeeds; otherwise failure;
-

Candidate sizes are evaluated from largest to smallest. The selected operating point is the largest K satisfying (10). This criterion distinguishes nominal symbol capacity $\log_2 K$ from the net payload rate after encryption, framing, and error correction.

F. Computational complexity

Let N be the index size, $A = |\mathcal{T}_{\text{cal}}|$, d the descriptor dimension, K the alphabet size, L the cover count, and F_θ the encoder inference cost. Clean embedding costs $O(NF_\theta)$. Multi-attack qualification is the dominant offline term, $O(ANF_\theta + AN \log K)$ with binary search over sorted boundaries. PCA fitting costs $O(Nd^2)$; sorting projected values $O(N \log N)$. Online cover retrieval and keyed mapping are $O(L)$ with pre-built buckets; Reed-Solomon decoding adds a fixed block cost. Offline qualification is intentionally amortised across subsequent transmissions.

Fig. 2 separates the offline index-construction phase from online protected transmission and recovery. Algorithm 1 summarises the transmission and recovery steps.

V. EXPERIMENTAL PROTOCOL

A. Datasets and partitions

The principal evaluation uses BOSSBase 1.01 [29], containing 10,000 grayscale photographs at 512×512 pixels. For each seed in $\{11, 29, 47, 71, 101\}$, a deterministic permutation provides 2,000 training images and a disjoint 8,000-image cover index.

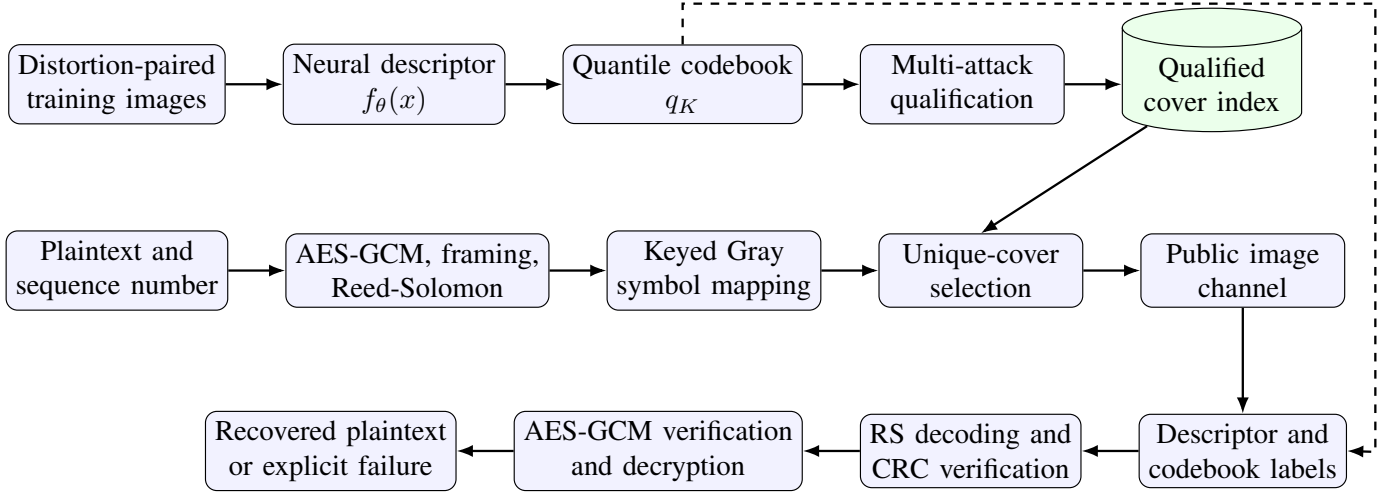


Fig. 2. NARCIS construction and transmission pipeline. Index construction and qualification are performed offline. The transmitted objects are unchanged images selected from the qualified index.

Cross-dataset validation uses Caltech-101 [30]: 9,144 natural RGB images from 101 object categories with variable source resolutions, centre-fitted to 256×256 for batching. Each of the same five seeds assigns 1,500 images to descriptor training and 7,000 disjoint images to the shared index. Semantic labels are used neither in training nor in codebook construction.

CIFAR-100 [31] is retained only for the controlled descriptor-comparison ablation (Section V-C): three seeds, 2,000 training images, an 8,000-image index, three messages per seed.

The encoder is trained separately on each partition for five epochs with AdamW, batch size 16, learning rate 10^{-3} , and weight decay 10^{-5} . BOSSBase evaluates ten eight-byte plaintexts per seed; Caltech-101 evaluates five. Reed-Solomon coding uses 128 parity bytes. Symbol repetition is set to one; redundancy is instead provided by byte-level parity, which avoids multiplying the required 348-464 cover sequences. The two principal campaigns comprise 75 protected messages and 1,575 message-condition trials.

B. Channel conditions and metrics

The 12 calibration transformations are those in (7). Eight holdout strengths are evaluated only at test time: Gaussian noise standard deviations 9 and 15; blur radii 1.2 and 1.8; crop fractions 0.08 and 0.12; and rotations of 5° and 9° . Together with the clean channel, this yields 21 conditions.

Two metrics are reported: (i) *symbol accuracy*, the fraction of transmitted covers assigned to their intended labels after channel processing; (ii) *complete-message success*, which is 1 only when error correction, CRC verification, authenticated decryption, and plaintext equality all succeed.

Confidence intervals are two-sided 95% Student-*t* intervals over the five seed-level values. They quantify variation across deterministic partitions and do not imply coverage over datasets or untested channels.

C. Baselines, descriptor comparison, and ablation

Local descriptor baselines. A normalised 16×16 low-frequency DCT descriptor and a normalised 64-bin grayscale histogram are subjected to the same quantile partition, calibration attacks, candidate codebook sizes, and protected-message feasibility rule as NARCIS. Neither descriptor receives invariance training. Under these conditions, neither the DCT nor the histogram formed a feasible codebook at *any* candidate alphabet size ($K \in \{64, 32, 16, 8, 4, 2\}$) in any of the three CIFAR-100 partitions. This outcome demonstrates that VICReg-style invariance training is a necessary prerequisite for the multi-attack qualification stage to produce a non-empty feasible codebook, not merely a preference that improves capacity.

Qualification ablation. The main ablation removes attack-based cover qualification while retaining the same neural descriptor, codebook size, protected messages, and decoder. It is applied to two settings: (a) the three-seed CIFAR-100 protocol, where it is computationally efficient to measure across all conditions, and (b) BOSSBase seed 11 with a matched 10-message, 21-condition protocol to confirm the BOSSBase result directly (Section VI-G).

Repetition factor check. A second check confirms the configured repetition factor of one.

D. Selection detectability

A logistic regression on seven global image statistics (mean, standard deviation, two gradient magnitudes, entropy, and two intensity quantiles) distinguishes selected from non-selected index images using a stratified 65/35 train/test split. A second detector extracts 52 high-pass residual co-occurrence summaries inspired by rich-model steganalysis [7] and applies an ExtraTrees ensemble. A third detector is a lightweight selection CNN trained for eight epochs on 64×64 views. The AUC measures detectability for all three. To assess the stability of the CNN estimate on the most sensitive partition (Caltech-101 seed 47), 20 independent training runs are performed.

TABLE II
SELECTED BOSSBase OPERATING POINTS ON 8,000-IMAGE INDICES.

Seed	K	bpc	Stable	Min. bucket	Median bucket
11	16	4	2,823 (35.3%)	92	164
29	16	4	2,430 (30.4%)	61	154
47	8	3	3,898 (48.7%)	177	510
71	16	4	2,687 (33.6%)	80	155
101	16	4	2,846 (35.6%)	99	146

TABLE III
SELECTED CALTECH-101 OPERATING POINTS ON 7,000-IMAGE RGB INDICES. ALL FIVE PARTITIONS SELECTED $K=16$.

Seed	Stable	Min. bucket	Median bucket	Max. bucket
11	1,678 (24.0%)	38	104	203
29	1,540 (22.0%)	31	82	185
47	2,432 (34.7%)	76	160	218
71	2,289 (32.7%)	45	110	284
101	1,895 (27.1%)	35	89	214

E. Sensitivity and projection analysis

A targeted reduced experiment (seed 11, 500 training images, 1,500-image index, three epochs, 96×96 inputs) compares descriptor dimensions 32, 64, and 128; three loss-weight configurations; and projections on the first three principal directions plus a fixed random direction.

A separate seed-11 full-protocol analysis evaluates 16 PCA directions (PC1-PC16) and 32 independent random projections to characterise the relationship between descriptor variance and stability under calibration attacks.

VI. RESULTS

A. Codebook feasibility

Table II reports the selected BOSSBase configurations. Four partitions supported 16 symbols (4 bits per cover); seed 47 supported eight symbols (3 bits per cover). Between 30.38% and 48.73% of the index remained stable under all 12 calibration attacks. Table III reports the Caltech-101 configurations; all five partitions selected $K=16$.

Each eight-byte plaintext required 348 covers at 4 bits per cover and 464 at 3 bits per cover. The measured net plaintext rates were 0.184 and 0.138 bits per transmitted cover, including encryption, framing, and error correction.

B. End-to-end robustness

All 1,050 BOSSBase trials succeeded across five seeds, ten messages, and 21 conditions. Mean symbol accuracy was 98.74%. The two most difficult holdout conditions were crop 12% (84.37%) and rotation 9° (91.75%); the worst individual accuracy was 81.61%. Reed-Solomon corrected at most 61 byte positions per message block, and authenticated decryption recovered every plaintext. The clean channel, both JPEG settings, both resize settings, calibrated crops, both blur calibration settings, and rotations up to 3° produced 100% mean symbol accuracy.

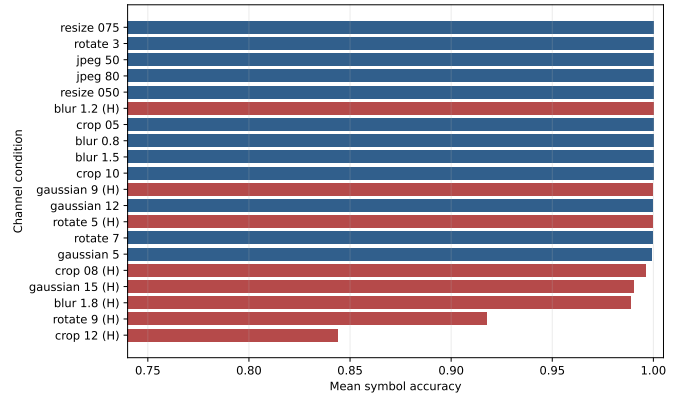


Fig. 3. Mean BOSSBase symbol accuracy over five seeds and ten messages per seed. Holdout strengths were excluded from codebook qualification. Complete-message success was 100% for every condition.

TABLE IV
PAYLOAD SCALABILITY AT $K=16$ (4 BITS PER COVER). AES-GCM OVERHEAD: 12-BYTE NONCE + 16-BYTE TAG. FRAMING: 10 BYTES. TWO RS BLOCKS ARE REQUIRED FOR 128-BYTE PAYLOADS.

Payload (bytes)	AES-GCM (bytes)	RS blocks ($\times 128$ parity)	Coded (bytes)	Covers required	Net bpc
8	36	1	174	348	0.184
32	60	1	198	396	0.646
128	156	2	422	844	1.213

Fig. 3 shows per-condition means for BOSSBase. All 525 Caltech-101 message-condition trials also recovered the authenticated plaintext. Mean symbol accuracy was 98.59%; the worst individual accuracy was 79.89% under the unseen 12% crop; Reed-Solomon corrected at most 63 byte positions. Fig. 4 compares the two datasets condition by condition.

C. Payload scalability

Table IV reports the implemented overhead breakdown and net throughput at $K=16$ for 8-, 32-, and 128-byte plaintexts. AES-GCM adds a 12-byte nonce and 16-byte authentication tag; protocol framing adds 10 bytes. For 8- and 32-byte payloads, a single Reed-Solomon block with 128 parity bytes suffices; 128-byte payloads span two blocks, doubling the effective parity to 256 bytes. Net throughput therefore scales sub-linearly with overhead: from 0.184 bits per cover at 8 bytes to 0.646 at 32 bytes and 1.213 at 128 bytes. Fig. 5 illustrates this scaling. The short-payload regime evaluating 8-byte messages in the principal campaigns represents an extreme overhead case; longer practical payloads achieve meaningfully higher efficiency under identical cryptographic and coding settings.

D. Training convergence

Fig. 6 reports the five-epoch training histories for BOSS-Base. The final loss values ranged from 0.066 (seed 47) to 0.157 (seed 29), a $2.4\times$ spread that reflects variation in partition character rather than training instability. Individual curves are not monotonic because each epoch samples new distortion pairs.

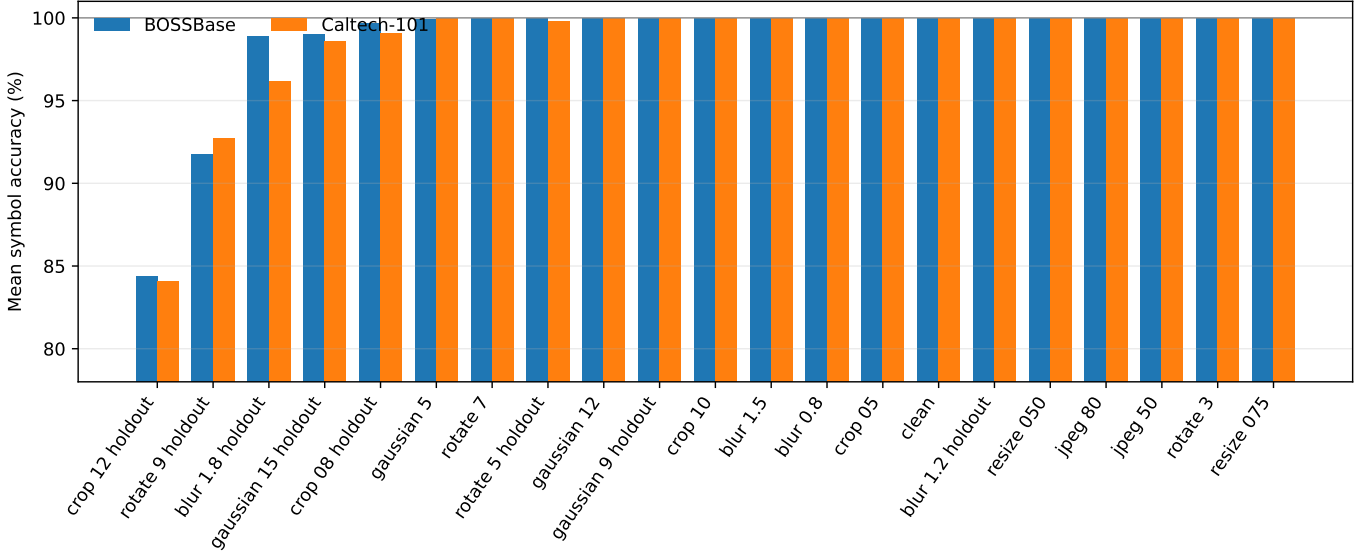


Fig. 4. Mean symbol accuracy across five BOSSBase and five Caltech-101 partitions. Calibration and holdout conditions use the same declared strengths. All 1,575 complete-message trials succeeded.

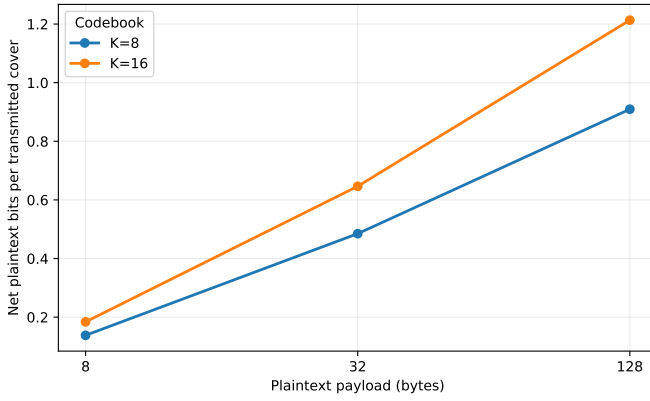


Fig. 5. Net plaintext throughput (bits per cover) as a function of payload size at $K=16$. Overhead (AES-GCM, framing, RS parity) is amortised across larger payloads; the 128-byte point uses two RS blocks with 128 parity bytes each.

TABLE V
SELECTION-DETECTION AUC (MEAN AND 95% STUDENT- t CI OVER FIVE PARTITIONS). INTERVALS EXCLUDING 0.5 INDICATE A MEASURABLE SHIFT.

Detector	BOSSBase	Caltech-101
Global logistic	0.518 [0.489, 0.547]	0.531 [0.504, 0.557]
Residual ExtraTrees	0.514 [0.486, 0.543]	0.533 [0.504, 0.562]
Selection CNN	0.515 [0.463, 0.566]	0.530 [0.481, 0.579]

E. Selection detectability

Table V reports the mean AUC and 95% Student- t confidence intervals over five partitions for each detector and dataset.

On BOSSBase, all three confidence intervals include 0.5. On Caltech-101, the two classical detectors show a small dataset-specific selection shift (intervals exclude 0.5); the CNN five-partition interval is wider and includes 0.5.

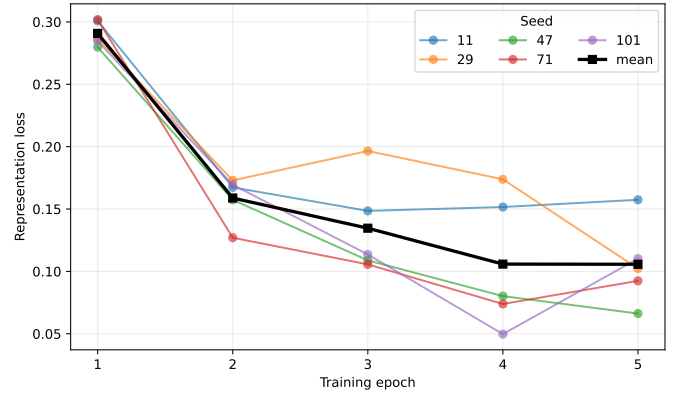


Fig. 6. BOSSBase training loss for the five deterministic partitions. Non-monotonicity reflects fresh distortion-pair sampling each epoch.

To assess whether the CNN result is stable on the most sensitive partition, 20 independent training runs were performed on Caltech-101 seed 47. Fig. 7 plots the resulting AUC distribution. The 20-run mean is 0.533 (95% CI [0.517, 0.548]), which excludes 0.5. This confirms a small but reproducible partition-specific selection signal on seed 47; Section VII-B places this finding in context.

F. Sensitivity, projection analysis, and complexity

Configuration sensitivity. Table VI summarises the reduced sensitivity experiment. Equal loss weights retained fewer stable covers and produced the smallest minimum bucket. Dimension 128 improved retained stability in this subset, but a confirmatory two-seed full-protocol run recovered 419/420 trials versus 420/420 for the corresponding 64-dimensional runs. The 64-dimensional configuration is therefore retained.

Projection candidates. The sensitivity study used PC1 as the default projection direction. Table VII compares PC1 with

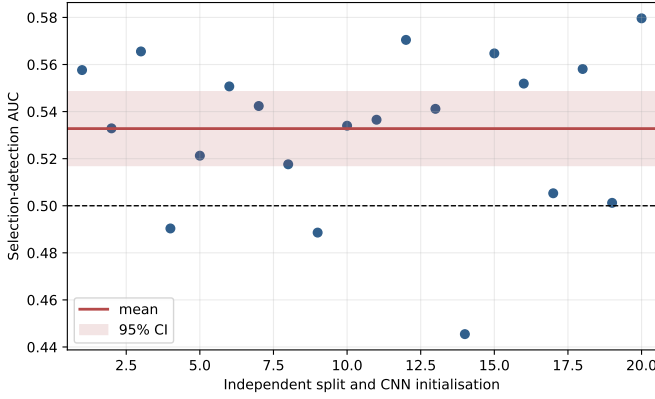


Fig. 7. Distribution of selection-CNN AUC over 20 repeated independent training runs on Caltech-101 seed 47. The 20-run 95% CI [0.517, 0.548] excludes chance performance; this is a partition-specific signal, not a protocol-level vulnerability.

TABLE VI
TARGETED SENSITIVITY STUDY ON THE REDUCED BOSSBase SUBSET
(SEED 11, 1,500-IMAGE INDEX, 96×96 INPUTS).

Configuration	Stable fraction	Min. bucket
64D, 25/25/1	0.335	9
32D, 25/25/1	0.375	19
128D, 25/25/1	0.412	20
64D, 1/1/1	0.284	5
64D, 25/10/1	0.357	19

PC2 and a survey of random directions on the BOSSBase seed-11 full-protocol index (8,000 images, $K=16$). PC1 and PC2 retain the same number of stable images but PC2 produces a higher minimum bucket (115 vs. 92). Among 32 randomly sampled fixed directions, the median minimum bucket is 101 and the best observed is 161 (random_15). A calibration-locked search that selected the direction with the highest minimum bucket before holdout evaluation achieved minimum bucket 161 and recovered 210/210 trials on seed 11. This single-partition result confirms the direction analysis but does not replace the five-seed PC1 evaluation. Fig. 8 plots minimum bucket against stable fraction for all evaluated directions.

Complexity. Native-image descriptor extraction averaged 87.83 s for 8,000 covers (99.85 images/s). Calibration averaged 1,246.10 s per partition. Complete evaluation averaged 754.63 s for 210 message-condition trials (3.59 s per trial).

G. Qualification ablation

CIFAR-100 controlled ablation. The three-seed CIFAR-100 experiment recovered 189/189 protected message-condition trials at $K=16$ (mean symbol accuracy 98.47%). Removing attack qualification reduced complete-message success from 100% to 43.65%.

BOSSBase seed-11 matched ablation. Table VIII reports the per-condition results of the matched BOSSBase seed-11 ablation (10 messages, 21 conditions). With qualification (pc1-qualified), all 210/210 trials succeeded. Without qualification (pc1-no-qualification), 142/210 trials succeeded (67.6%). Crop

TABLE VII
PROJECTION CANDIDATES ON BOSSBase SEED-11 ($K=16$, 8,000-IMAGE INDEX). PC1 AND PC2 RETAIN THE SAME STABLE IMAGES (2,823) BUT DIFFER IN BUCKET BALANCE. THE CALIBRATION-LOCKED RANDOM DIRECTION (BEST OF 32 TRIALS) ACHIEVES THE HIGHEST MINIMUM BUCKET.

Direction	Expl. var.	Stable images	Min. bucket
PC1	94.07%	2,823 (35.3%)	92
PC2	5.19%	2,823 (35.3%)	115
Random, median of 32	—	3,630 (45.4%)	101
Random, best of 32	—	3,867 (48.3%)	161

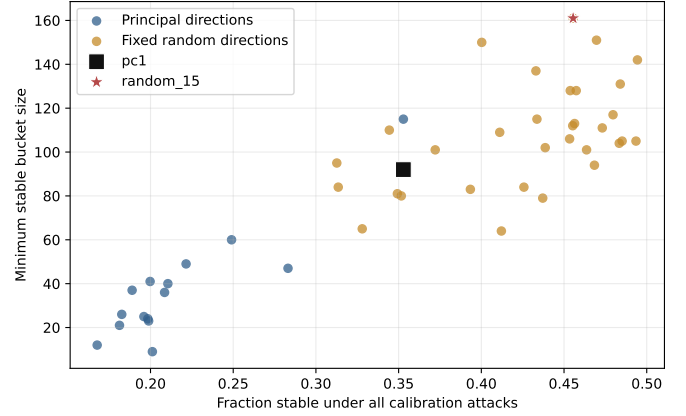


Fig. 8. Minimum bucket size vs. stable fraction for PCA directions (PC1-PC16) and 32 random projections on BOSSBase seed-11. Random directions consistently dominate the PCA directions in minimum-bucket size, motivating attack-aware projection selection as a future extension.

conditions failed completely (symbol accuracy 38-63%, far beyond the Reed-Solomon correction capacity of 64 bytes). Rotation conditions at $\geq 7^\circ$ also failed outright. This confirms that invariance training alone is insufficient: qualification is the dominant robustness mechanism for distortions with sharp spatial structure (crops, large rotations).

H. Position relative to published methods

Table IX separates directly measured NARCIS properties from published context. Abdulsattar [11] and WYSAWIS [12] provide high nominal intra-image or category-based representations without end-to-end protocol evaluation. Guo and Ping [6] report higher nominal capacities and high average robustness on larger datasets, with a capacity advantage at their operating points. NARCIS's supporting position is protocol completeness, not maximum nominal capacity.

VII. DISCUSSION

A. What the experiments establish

Four conclusions are supported. First, stable descriptor averages are insufficient for communication: the multiplicity of every required symbol must be verified. Second, cover qualification is the dominant robustness component: removing it causes catastrophic failure on cropping and large-rotation conditions that the neural descriptor alone cannot tolerate. Third, forward error correction allows message recovery at

TABLE VIII

BOSSBASE SEED-11 MATCHED ABLATION: COMPLETE-MESSAGE SUCCESS OVER 10 MESSAGES PER CONDITION (21 CONDITIONS, 210 TRIALS TOTAL). “QUALIF.” = PC1-QUALIFIED (NARCIS STANDARD); “NO QUALIF.” = PC1 WITHOUT QUALIFICATION. (H) = HOLDOUT STRENGTH.

Condition	Qualif.	No qualif.	Mean sym. acc. (no qual.)
Clean	10/10	10/10	1.000
JPEG 80, 50	10/10	10/10	>0.993
Gaussian σ 5, 12	10/10	10/10	0.926-0.930
Gaussian σ 9 (H), 15 (H)	10/10	10/10	0.925-0.926
Blur 0.8, 1.5	10/10	10/10	0.872-0.950
Blur 1.2 (H), 1.8 (H)	10/10	10/10	0.847-0.904
Resize 0.75, 0.50	10/10	10/10	0.936-0.959
Crop 5%, 10%	10/10	0/10	0.430-0.627
Crop 8% (H), 12% (H)	10/10	0/10	0.381-0.524
Rotate 3°	10/10	10/10	0.887
Rotate 7°	10/10	0/10	0.719
Rotate 5° (H)	10/10	2/10	0.788
Rotate 9° (H)	10/10	0/10	0.666
Total	210/210	142/210 (67.6%)	—

TABLE IX

POSITIONING OF NARCIS. VALUES FROM DIFFERENT PUBLICATIONS ARE NOT TREATED AS A COMMON-PROTOCOL RANKING. (H) = HOLDOUT.

Property	Abdulsattar [11]	WYSAWIS [12]	Guo-Ping [6]	NARCIS
Representation	Eigenvalue comparisons in overlapping blocks	15-bit block hashes, 16 category sequences	PZM features, stability-aware quantisation and clustering	Learned invariant descriptors with quantile codebook
Primary capacity	High nominal single-cover capacity	High nominal block/category capacity	10-15 bits on Holidays, VOC, ImageNet	3-4 bits on BOSSBase; 4 bits on Caltech-101
Multi-attack qualification	Not evaluated	Not evaluated	Yes	Yes
Holdout attack strengths	Not evaluated	Not evaluated	Not established from cited summary	8 holdout strengths, 21 total conditions
Message-level error correction	Not reported	Not reported	Not stated	Reed-Solomon; worst case 63 corrections
Authenticated payload + metadata	Not reported	Not reported	Not stated	AES-GCM; replay guard; identity binding
Net throughput range	—	—	Not reported	0.184-1.213 bpc (8-128 bytes at $K=16$)
Measured complete recovery	Not reported	Not reported	Not directly comparable	1,575/1,575 trials

symbol accuracies well below 100%. Fourth, cryptographic verification changes the evaluation criterion: a trial succeeds only when the exact plaintext is authenticated and recovered.

The five-seed results quantify finite-index and cross-corpus variability. The same rules yield a feasible 16-symbol alphabet on four BOSSBase partitions and an eight-symbol alphabet on one, while all five Caltech-101 partitions support 16 symbols. Reporting the lower BOSSBase operating point avoids selection of only favourable partitions.

B. Security interpretation

Propositions 1 and 2 establish payload confidentiality and metadata authenticity under AES-GCM. These properties hold independently of whether selection is detectable.

The AUC experiments address a different question: whether the cover selection policy creates a statistically detectable distribution shift. The Caltech-101 classical detectors show mean AUCs of 0.531 and 0.533 with confidence intervals that

exclude 0.5, indicating a weak but real selection signal. The 20-repeat CNN result on the most sensitive partition (mean 0.533, CI [0.517, 0.548]) confirms this signal is reproducible.

These findings must be interpreted carefully. The shift is measured against the full un-qualified index, not against an external cover database. A warden who has access to the same index can therefore use this signal. A warden who does not know the index must estimate the reference distribution from publicly available images, which weakens the detection advantage. The experiment establishes that the qualification policy introduces a detectable selection leak on Caltech-101 and motivates future work on distribution-preserving selection strategies.

C. Projection selection and the PC1 choice

Table VII reveals a counter-intuitive finding: PC1 (the direction of maximum descriptor variance) yields the lowest minimum bucket (92) among all evaluated projections on

BOSSBase seed 11. PC2 achieves the same stable-image count but a higher minimum bucket (115). The 32 random directions achieve substantially higher stable fractions (median 45.4%) and higher median minimum bucket (101), with the best reaching 161.

This pattern arises because PC1 captures the descriptor dimensions with the highest variance, and high-variance dimensions encode precisely the visual content that is most perturbed by the channel attacks. The calibration attacks therefore cause the largest boundary-crossing rates along PC1. Lower-variance or random directions encode less channel-sensitive content and retain more stable covers as a result.

PC1 is retained in the principal evaluation because it is deterministic, variance-maximising, and reproducible without additional calibration data. A calibration-locked projection (random_15, minimum bucket 161) recovered 210/210 trials on seed 11, confirming the gain is realisable. However, this single-partition result does not replace the five-seed PC1 campaign, and an attack-aware direction trained on calibration data while locked before holdout evaluation is identified as the highest-priority future extension.

D. Reed-Solomon parameter choice

The choice of 128 parity bytes per Reed-Solomon block is conservative but verifiable. The maximum observed correction across all 1,575 trials was 63 byte positions (Caltech-101), leaving a 1-byte margin to the theoretical capacity of $\lfloor 128/2 \rfloor = 64$ correctable errors. The 128-byte payload evaluation uses two RS blocks, each with 128 parity bytes, maintaining the same per-block correction capacity while doubling the total parity overhead.

A tighter parameter (e.g., 64 parity bytes, $t=32$ corrections) would halve the cover sequence length for short payloads but would fail to correct the worst observed conditions (BOSSBase seed 47 under 12% crop: 61 corrections; Caltech-101 seed 29 crop 12%: 63 corrections). The current choice is therefore the minimum that supports the tested attack range with a measurable but non-zero margin.

E. Comparative context

Under the same quantile partition, attack set, candidate sizes, and protected-message demand, neither the 16×16 DCT descriptor nor the 64-bin histogram formed a feasible codebook at any tested alphabet size. This is not a result about DCT and histogram methods in general; it demonstrates that the VICReg invariance objective is a prerequisite for the multi-attack qualification stage to produce a usable codebook under the 12-attack calibration protocol. Without invariance training, too few covers survive all calibration conditions to satisfy the multiplicity constraint at any alphabet size.

Guo and Ping's stability-regularised clustering [6] shares the objective of retaining stable covers but reports robustness at the descriptor level on larger datasets with higher nominal capacities. A direct comparison requires matching datasets, attack definitions, and protocol overheads. The present results are complementary rather than competitive: they characterise the operating regime where the feasibility constraint is the

binding factor, whereas Guo-Ping target maximum nominal capacity at high average robustness.

F. Limitations

The evaluation has four main limitations:

- 1) Caltech-101 adds variable-resolution RGB content, but its channel input is centre-fitted to 256×256 for batching. Evaluation on larger native-resolution colour corpora would test a broader regime.
- 2) The local baselines (DCT, histogram) isolate descriptor feasibility; they are not faithful reimplementations of complete published coverless systems.
- 3) The learned selection detector is a lightweight CNN designed for the selected-vs.-non-selected task, not an SRNet reproduction. Traffic-level sequence analysis is also outside this study.
- 4) PC1 is reproducible and variance-maximising but is not stability-optimal. Attack-aware projection selection is identified as the highest-priority future extension.

VIII. CONCLUSION

NARCIS integrates learned image descriptors, balanced codebooks, attack-based cover qualification, keyed assignment, authenticated encryption, replay control, and Reed-Solomon correction into one selection-based coverless protocol. It recovered 1,050/1,050 protected message-condition trials on BOSSBase and 525/525 on RGB Caltech-101, including eight holdout attack strengths. BOSSBase supported 3-4 bits per cover; all Caltech-101 partitions supported 4 bits per cover. Mean symbol accuracies were 98.74% and 98.59%.

Net throughput at $K=16$ scales from 0.184 bits per cover for 8-byte payloads to 0.646 for 32-byte and 1.213 for 128-byte payloads, demonstrating that the short-payload overhead is not a structural limitation of the protocol. A matched BOSSBase seed-11 ablation confirmed that removing qualification reduces complete-message success from 210/210 to 142/210 (67.6%), with crop and large-rotation conditions failing completely. A calibration-locked projection recovered 210/210, motivating attack-aware direction selection as the highest-priority extension.

Formal propositions ground the cryptographic protections in AES-GCM's established security properties. Detectability is reported transparently: BOSSBase shows no significant signal; Caltech-101 shows a reproducible partition-specific CNN signal (0.533, CI [0.517, 0.548]) that identifies a selection leak attributable to the qualification filter rather than to the protocol's cryptographic components.

Three extensions follow from observed limits. First, attack-aware projection selection should be trained on calibration data while locked before holdout evaluation. Second, executable published-protocol baselines should be evaluated under the same payload protection, attacks, and finite-index demand. Third, larger native-resolution colour corpora and traffic-level sequence detectors should extend the present image-level analysis.

ACKNOWLEDGMENTS

This work was supported by the authors' institutions. No external financial support was received.

APPENDIX

DATA AND CODE AVAILABILITY

BOSSBase 1.01 is publicly available at <https://agents.fel.cvut.cz/boss/>; Caltech-101 at <https://data.caltech.edu/records/mzrjq-6wc02>; CIFAR-100 at <https://www.cs.toronto.edu/~kriz/cifar.html>. The NARCIS implementation, attack definitions, evaluation scripts, fixed seeds, consolidated CSV and JSON result tables, and manuscript sources are publicly available at <https://github.com/EkodeckStephane/NARCIS>. Trained checkpoints are not redistributed; the repository provides the commands to retrain.

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